

How to Run a Kubernetes Cluster Locally with Kind

Kind is a tool used for running Kubernetes (k8s) locally using Docker container 'nodes'. Its primary purpose is to test k8s, but it can also be used in your CI system or for local development.



Kind serves various purposes in the Kubernetes ecosystem depending upon your use case. It can be used to learn Kubernetes, to make sure that your Kubernetes application manifests are correct, or to test your deployments locally. In this article, we will see how to use Kind to create a Kubernetes cluster locally and run a Web application using it.

Let's invoke the Kind executable to see what are the available options:

```
> kind
kind creates and manages local Kubernetes clusters using
Docker container 'nodes'
```

Usage:

```
kind [command]
```

Available Commands:

```
build      Build one of [node-image]
completion Output shell completion code for the specified
shell (bash, zsh or fish)
create     Creates one of [cluster]
delete     Deletes one of [cluster]
export     Exports one of [kubeconfig, logs]
get        Gets one of [clusters, nodes, kubeconfig]
help       Help about any command
load       Loads images into nodes
version    Prints the kind CLI version
```

Flags:

```
-h, --help          help for kind
--loglevel string    DEPRECATED: see -v instead
-q, --quiet          silence all stderr output
-v, --verbosity int32 info log verbosity
--version           version for kind
```

Use "kind [command] --help" for more information about a command.

Cluster creation and interaction

Now let's create a Kubernetes cluster locally and interact with it:

```
> kind create cluster
Creating cluster "kind" ...
> Ensuring node image (kindest/node:v1.21.1) >
> Preparing nodes >
> Writing configuration >
> Starting control-plane >
> Installing CNI >
> Installing StorageClass >
Set kubectl context to "kind-kind"
```

You can now use your cluster with:

```
kubectl cluster-info --context kind-kind
```

Not sure what to do next? Check out <https://kind.sigs.k8s.io/docs/user/quick-start/>.

The above command will bootstrap a k8s cluster using a pre-built node image. In fact, once the above command completes, you can use Docker to verify that a new image has indeed been downloaded:

```
> docker images | rg kind
kindest/node                                     <none>
32b8b755dee8   8 months ago   1.12GB
```

The default cluster name is *kind*. And you can confirm it using the following command: